

A Packing of 13 Four-Spaces in $\mathbb{R}^9$

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October 9, 2026

Abstract

We give 13 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.407101. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.392880, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 13 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.407101 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 13$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.392880, obtained by deleting 1 of the 14 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 13/12 = 2.407407$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N13.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
120355063330612121592725085433619877117600247149688832890866523206031334
304242060077755180586611395186331388569918672561289215574836059995644035
8000222379217808890991672906739969989507956637399
```

denominator:

```
500000000000614669073304355776480310801550242028511440434606497579964032
305880636052574350024891023560351910245178941014094301518652866260680037
015180786441172319898429480746856487969717903523
```

B Coordinates

13 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N13.txt`.

```
-98096102223 93851004525 -615724639330 604017186729651804536533 194554832996 85705036725 450847157602250966076771 -102728623152 471691538376 -71423711504
-663597627523 443556847453 297990397086 36987391545237780158059 -489785346352 -284685932685 2230961212365591206534 138106237849 -4832222012 339907579107
387608555851 231952434782 -76655498146 -28236634593317953038334 -371724077853 -9727753373 -8057406237364680894083 -230177372612 710525816440 -182905114837
-217962791620 -190256546626 225611134723 3111637177455141599227 -57628412971 -352804550990 5510334887837040345356 627181759907 -76740014061 -480748363674
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-212636426334 188157648153 243673047531 -216913775336845443070 -170115188395 -327230758172 7295473115548487413414 377500670929 257465417258 84015079026
225478483824 -48602069184 -13614683163 -5448542961729091482307 -524466618028 88018375727 46268125479942733209574 384762573953 412826420138 262165721754
-211015601421 -655371213526 -79791998931 416421476733557469560 742940458573 199987463610 165346224720 323233280266 -248793781809 -215315941779 178526698200
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```